

- Introduction

This project involved the development of a K-Means cluster dataset hosted on the Registry of Open Data on AWS. The speed, and precipitation, using Python and PyCharm as the that may not be immediately visible through basic analysis

```
[17]: import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

```
•[19]: # Load the dataset
df = pd.read_csv("Weather_Data.csv")
df.head()
```

```
[19]:
```

	Date/Time	Temp_C	Dew Point	Temp_C	Rel Hum_
0	1/1/2012 0:00	-1.8		-3.9	
1	1/1/2012 1:00	-1.8		-3.7	
2	1/1/2012 2:00	-1.8		-3.4	
3	1/1/2012 3:00	-1.5		-3.2	
4	1/1/2012 4:00	-1.5		-3.3	

```
•[21]: # Select relevant numeric columns and drop any mi
features = ['Temp_C', 'Rel Hum_%', 'Wind Speed_kmh']
df_clean = df[features].dropna()
```

```
•[23]: # Standardize the data
scaler = StandardScaler()
scaled_data = scaler.fit_transform(df_clean)
```

```
•[25]: # Apply K-Means clustering
kmeans = KMeans(n_clusters=3, random_state=42)
```